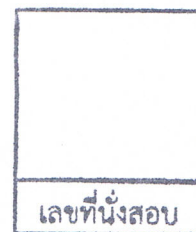


College of Industrial Technology
King Mongkut's University of Technology North Bangkok



Final Examination of Semester 1

Year: 2012

Subject: 392153 Chemistry III

Section: 15-16

Date: 27 September 2012

Time: 10.00-12.000

Name: _____ ID: _____ Field of Study: _____

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks are not allowed.
4. Dictionaries are NOT permitted.
5. A calculator is permitted.
6. No any electronic communication devices allow in the exam room.
7. The examination has 10 pages (including this page and equations), total scores are 70 points.
8. Write all your answers on this exam.

$$k = A \cdot \exp(-E_a/RT)$$

$$R = 8.314 \text{ J/K}\cdot\text{mol}$$

$$\ln(k_2/k_1) = -E_a/R \left((1/T_2) - (1/T_1) \right)$$

$$\ln k = -\frac{E_a}{R} \frac{1}{T} + \ln A$$

Rate Law

Concentration-Time
Equation

$$\text{rate} = k [A]$$

$$\ln[A] = \ln[A]_0 - kt$$

$$\text{rate} = k$$

$$[A] = [A]_0 - kt$$

$$\text{rate} = k [A]^2$$

$$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$$

1. A reaction is 50% complete in 10 minutes. It is allowed to proceed another 5 minutes. How much of the reaction would be complete at the end of these 15 minutes if the reaction follows zero order kinetics? Let $[A]^0$ be $1 \times 10^3 \text{ mol dm}^{-3}$

Total points = 10 points

2. Determine the rate law for the reaction $2\text{NO} + 2\text{H}_2 \rightarrow \text{N}_2 + 2\text{H}_2\text{O}$ from the following kinetic data:

Experiment	Partial pressure of H_2	Partial pressure of NO	Initial rate (torr/sec)
1	400	359	9.0
2	400	300	6.2
3	400	152	1.5
4	289	400	9.6
5	205	400	6.6
6	147	400	4.7

(2.1) Use the below data to determine the order of the reaction with respect to

1 Nitrogen monoxide

2 Hydrogen

(2.2) Write a rate equation for the above reaction

(2.3) Calculate the rate constant, giving its units

Total points = 15 points

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3. Determine the activation energy for the decomposition of nitrogen pentoxide from the following data:

Experiment	Temperature (K)	k (min ⁻¹)
1	273	0.0000472
2	298	0.00203
3	338	0.292

Total points = 10 points

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4. Why would the initial reaction rate be expected to increase by increasing each of the following parameters? Answer by TRUE or FALSE the sentences .

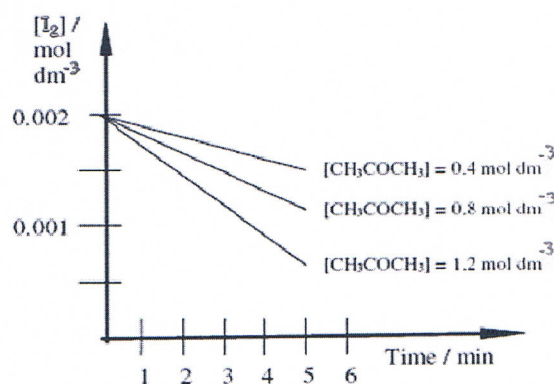
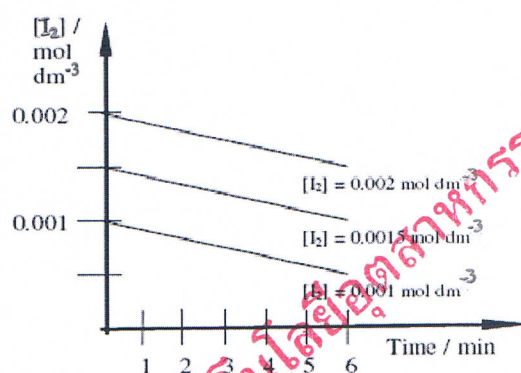
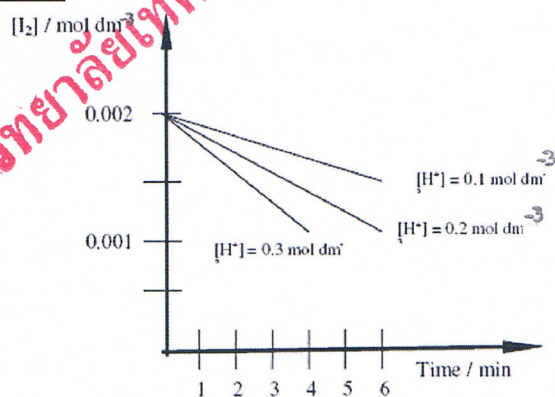
(Total points = 5 points)

Answer	Questions
	An increase in pressure of a gaseous reaction
	Gaseous reactant particles are brought further.
	Number of gaseous reactant particles per unit volume increases.
	Number of effective collision decreases.
	Frequency of effective collision decreases.
	Rate of reaction is proportional to the frequency of effective collisions.
	An increase in temperature
	Number of reactant particles with $E \leq E_a$ increases.
	Number of effective collisions increases.
	Frequency of effective collisions decreases.
	Rate of reaction is not proportional to the frequency of effective collisions.
	Rate of reaction decreases.

5. Iodine and propanone (CH_3COCH_3) react together in an aqueous acidic solution according to the equation



The rate of the reaction can be measured by recording the reduction of the concentration of the iodine by the decrease in the intensity of its colour as measure in a colorimeter. Three sets of separate experiments were performed in which the initial concentration of each of the reactants, iodine, propanone and acid was varied in turn, the other two being kept constant. The results are shown below in graphical form.

Graph 1**Graph 2****Graph 3**

Use the graphs to determine the order of reaction with respect to

5.1 propanone

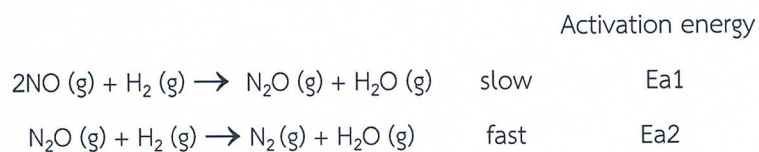
5.2 iodine

5.3 acid

Total points = 15 points

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6. At 700 °C nitrogen monoxide can be reduced by hydrogen via a two-step exothermic reaction as follows:



6.1 Write the rate equation for the above reaction, stating the units of the rate constant.

6.2 Sketch and label a reaction pathway diagram for the reaction, indicating clearly all relevant energy changes.

Total points = 15 points

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